MatchSense AI: A News-Aware, Lineup-Conditioned Forecasting System for the English Premier League

Architecture, Walk-Forward Evaluation, and a Critical Appraisal

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Abstract

MatchSense AI answers natural-language questions such as *"Will Manchester City win tonight?"*. It resolves the question to a verified fixture in the user's timezone and gathers 72 hours of news for both clubs. It turns that text into typed, sourced availability evidence, predicts both starting line-ups probabilistically, and adjusts a pre-existing recency-weighted Poisson score model through a separately estimated line-up layer, combined by Monte Carlo simulation. Its central design rule is that the language model *never* produces a probability. It only reads text, and every number comes from a fitted statistical model whose out-of-sample behaviour has been measured. We evaluate each component with leakage-controlled, chronological protocols. The baseline goal model beats a class-frequency prior on an untouched season (log loss 1.028 vs. 1.082), and a multinomial-logistic alternative adds nothing (selected blend weight 0%). A pooled logistic start-probability model is clearly useful (test log loss 0.400 vs. 0.461; top-11 precision 0.798 vs. 0.742). However, *line-up effects on goal rates could not be identified* from the one season of line-up data available. Player-level ridge effects that won on validation lost on test (up to +0.027 log loss, 95% CI $[+0.000, +0.053]$), and no team-level specification improved held-out log loss with an interval excluding zero. The deployed line-up layer is therefore an explicitly labelled, prior-dominated Bayesian model, and the unadjusted baseline is always shown beside it. We document three evaluation rounds, each changed after viewing test results, and conclude that the test folds are no longer a clean holdout. We then give a systematic appraisal of the system's strengths, flaws and threats to validity, including model artefacts found by replaying the system. Examples are that a goalkeeper's absence is attributed mainly to his own team's attack, and that a draw is never the most likely outcome.

*Independent technical report, November 2026. All numerical results are produced by the repository's own scripts and artifacts (Appendix A).

Keywords: football forecasting; Poisson regression; Dixon–Coles; line-up prediction; information extraction; large language models; Monte Carlo simulation; data leakage; proper scoring rules; negative results.

Contents

1 Introduction **3**

2 Related work **3**

3 System overview **4**

4 Data **6**

5 Methodology **6**
5.1 Baseline match model (unchanged) 6
5.2 Query and fixture resolution 6
5.3 Evidence extraction and corroboration 7
5.4 Start-probability (selection) model 7
5.5 Line-up impact model 7
5.6 Monte Carlo forecast and explanations 8

6 Experimental design **8**

7 Results **8**
7.1 Baseline: Poisson versus logistic 8
7.2 Start-probability model 9
7.3 Line-up impact candidates 9
7.4 Posterior of the deployed layer 9
7.5 Does line-up information help at all? 9
7.6 Calibration 9
7.7 Evaluation history and disclosure 10

8 Case study (demonstration data) **12**

9 Software validation **12**

10 Critical appraisal **12**
10.1 Strengths 12
10.2 Flaws and weaknesses 13
10.3 Pros and cons versus simpler alternatives 13
10.4 Threats to validity 13
10.5 Responsible use 13

11 Future work **15**

12 Conclusion **15**

A Reproducibility **18**

B Data availability **18**

1 Introduction

Statistical models of football results have a long history. It runs from independent Poisson goal models (Maher, 1982) and the low-score dependence correction of Dixon and Coles (1997), through rating systems such as Elo (Elo, 1978; Hvattum and Arntzen, 2010), to bivariate, Bayesian-hierarchical and dynamic formulations (Karlis and Ntzoufras, 2003; Rue and Salvesen, 2000; Baio and Blangiardo, 2010; Koopman and Lit, 2015) and machine-learned feature models (Hubáček et al., 2019; Bunker and Thabtah, 2019). These models are built from results and schedules. Much of the information that practitioners consider decisive before a match is not in a results table. It arrives as unstructured text: injury bulletins, suspensions, press-conference remarks, predicted and confirmed line-ups.

Large language models (LLMs) make such text easy to read and equally easy to misuse. An LLM asked "will City win?" produces a fluent answer with no calibrated probabilistic meaning and a well-documented tendency to hallucinate (Ji et al., 2023). Retrieval augmentation (Lewis et al., 2020) grounds the text the LLM sees, but it does not make the model's numerical claims calibrated. MatchSense is built around the opposite division of labour:

> *The LLM is an information-extraction and query-understanding component whose output schemas contain no probability fields. All probabilities come from fitted statistical models, and when the measured evidence for a component is weak, the system says so in every response.*

Contributions.

1. **An architecture** (Section 3) that layers news- and line-up-derived adjustments on top of an *unchanged*, independently validated baseline through an offset Poisson GLM, so the baseline always remains available for comparison.
2. **A leakage-controlled evaluation** (Sections 6 and 7) of the baseline, the selection model and twenty-two line-up-impact candidates, with matchday walk-forward folds and paired bootstrap intervals.
3. **A negative result, reported with its history** (Section 7.7). Line-up effects on goal rates are not identifiable from one season, and the most expressive models overfit. We disclose three post-hoc evaluation rounds.
4. **A critical appraisal** (Section 10): strengths, concrete flaws with severity ratings, a pros/cons comparison against simpler alternatives, threats to validity, and responsible-use considerations.

2 Related work

Goal-based models. Maher (1982) modelled home and away goals as independent Poisson variables with attack and defence parameters. Dixon and Coles (1997) added a dependence correction τ_ρ for scores of 0–0, 1–0, 0–1 and 1–1, and exponential time-weighting of past matches. Both ideas appear in our baseline as recency weights and a Dixon–Coles score grid. Karlis and Ntzoufras (2003) introduced bivariate Poisson models. Rue and Salvesen (2000) and Koopman and Lit (2015) let team strengths evolve over time, and Baio and Blangiardo (2010) placed the parameters in a Bayesian hierarchy, which shrinks weakly identified effects. That last idea is central to our handling of line-up effects.

Ratings and feature models. Elo ratings (Elo, 1978) are competitive predictors of football outcomes (Hvattum and Arntzen, 2010) and serve as inputs to our baseline. Feature-based learners, from gradient-boosted trees on relational features (Hubáček et al., 2019) to generic ML

pipelines (Bunker and Thabtah, 2019), typically gain little over well-specified goal models once proper scoring rules are used. Our walk-forward comparison of Poisson and multinomial-logistic models agrees (Section 7.1).

Player contribution. Regularised adjusted plus-minus estimates individual contributions by ridge regression (Hoerl and Kennard, 1970) on line-up indicators (Sill, 2010). Kharrat et al. (2020) adapted plus-minus to football and stressed heavy regularisation and multi-season data. Action-value models such as VAEP (Decroos et al., 2019) use event streams that are not available to us. Our failure to identify player effects from one season is consistent with these observations.

Forecast evaluation and leakage. We use strictly proper scoring rules (Gneiting and Raftery, 2007): logarithmic loss, the Brier score (Brier, 1950) and the ranked probability score (Epstein, 1969), which Constantinou and Fenton (2012) recommend for ordered football outcomes. Evaluation is rolling-origin (Tashman, 2000). Temperature scaling follows Guo et al. (2017). Selection-induced optimism when the same data guide model choice and performance estimation is analysed by Cawley and Talbot (2010), and the one-standard-error rule by Hastie et al. (2009). Leakage is formalised by Kaufman et al. (2012).

Sampling and uncertainty. Line-ups are sampled with the Gumbel-top-k trick (Kool et al., 2019), an approximation to conditional Bernoulli sampling (Chen et al., 1994). Coefficient uncertainty uses a Laplace approximation (Tierney and Kadane, 1986). Paired intervals use the bootstrap (Efron and Tibshirani, 1993). Near-duplicate news detection uses word shingles (Broder, 1997).

3 System overview

Figure 1 shows the request path. The system is a Python package (`src/matchsense/`, about 4,500 lines) served by FastAPI with a build-free JavaScript dashboard. It runs in two modes. *Live* mode uses public endpoints and optional API keys. *Demo* mode replays a recorded snapshot of the Fantasy Premier League (FPL) endpoints with *simulated* news, which is clearly labelled as such.

Table 1 maps the components to source files.

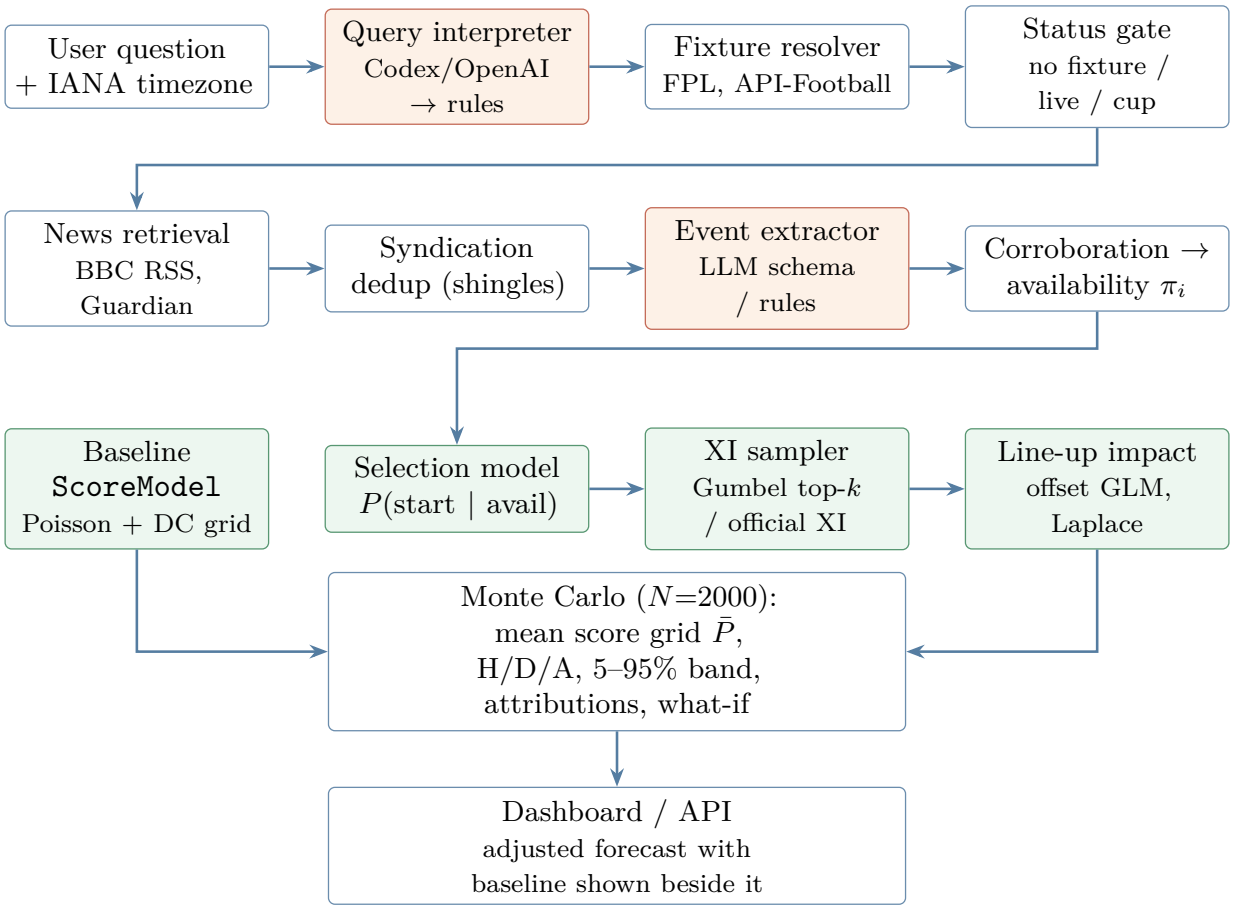


Figure 1: Request path. Orange boxes use an LLM, but only to emit *structure* (intent, clubs, date phrase, typed events). Green boxes are fitted statistical components that produce all numbers. Each LLM stage has a deterministic fallback.

Module	Responsibility
`interpreter.py`	LLM query understanding (Codex CLI → OpenAI API → rule parser); strict JSON schema; validated and cached per question.
`query.py`, `teams.py`	Club aliases, intent, timezone-aware windows ("tonight" = 17:00–06:00 local), what-if player phrases.
`providers/`	FPL, openfootball, API-Football, football-data.org, BBC/Guardian/NewsAPI/GNews; demo replay.
`http.py`	SQLite response cache, retries with back-off, per-host throttling, stale fallback.
`intelligence.py`	Syndication clustering, LLM/rule extraction (Pydantic), corroboration into availability beliefs.
`players.py`	Name-to-squad identity resolution (initials, accents, particles; refuses ambiguous matches).
`lineup_data.py`, `training.py`	Leakage-safe selection state; walk-forward folds; one-SE rule; bootstrap.
`impact.py`, `lineups.py`	Offset-Poisson line-up GLM (MAP + Laplace); selection model; constrained XI sampling.
`baseline.py`, `simulate.py`	Existing `ScoreModel` + season roll-forward; Monte Carlo score grids.
`pipeline.py`, `storage.py`, `api.py`	Orchestration, explanations, what-if, prediction history and scoring, HTTP API.

Table 1: Principal modules.

4 Data

Historical results. 9,880 completed Premier League matches (2000-01 to 2025-26) from the `openfootball/england` repository (openfootball contributors, 2026), ingested into a normalised PostgreSQL schema by the project's existing pipeline. The pipeline uses explicit team aliases, rejects duplicate and impossible scores, and stores a fingerprint and the raw source text for each match.

Historical line-ups. Only 2025-26 has full line-up files locally: 380 matches with eleven parsed starters per side, substitution minutes and scorers. This single season is the *only* training data for the line-up layer. Team-match observations with fewer than three prior matches that season are excluded (a "usual XI" is undefined). This leaves 700 of 760 observations.

Live sources. Current fixtures, results, squads, per-player starts, minutes and goals, and the league's structured availability flags (status, chance of playing, news text and timestamp) come from the public FPL endpoints (The Premier League, 2026). News comes from BBC Sport club RSS feeds and, with keys, the Guardian Open Platform, NewsAPI and GNews. API-Football and football-data.org add cup and European fixtures, injuries and official XIs. Items without a parseable publication time are dropped, because the 72-hour window cannot be enforced for them.

What is missing. There is no timestamped historical *news* archive for the evaluation season. The news-to-availability channel therefore cannot itself be backtested. Only the line-up channel it feeds can (Section 7.5).

5 Methodology

5.1 Baseline match model (unchanged)

The baseline is the repository's deployed `ScoreModel`. Home and away goals follow Poisson GLMs (McCullagh and Nelder, 1989),

$$y_h \sim \mathrm{Pois}(\lambda_h), \quad y_a \sim \mathrm{Pois}(\lambda_a), \qquad \log \lambda_h = \mathbf{x}^\top \boldsymbol{\beta}_h, \quad \log \lambda_a = \mathbf{x}^\top \boldsymbol{\beta}_a, \tag{1}$$

fitted with an L2 penalty $\alpha = 0.03$ and recency weights $w = 2^{-\mathrm{age}/1460\,\mathrm{d}}$. The 32 pre-match features $\mathbf{x}$ include Elo ratings and their difference, exponentially weighted scoring and conceding rates (overall and by venue), points and goal difference per game, five-match form, draw tendency, rest-day difference, promotion flags and matchday phase. Scores are mapped to a 21×21 Dixon–Coles grid

$$P(i,j) \propto \tau_\rho(i,j)\, \frac{\lambda_h^i e^{-\lambda_h}}{i!}\, \frac{\lambda_a^j e^{-\lambda_a}}{j!}, \tag{2}$$

where validation selected $\rho = 0$, i.e. independence. For live use the bundled end-of-2025-26 `SeasonState` is copied, the existing `start_season` transition is applied (regression to the mean and promoted-team priors), and completed 2026-27 results are replayed in date batches. A result counts as known only if kickoff + 2 h 30 min precedes the prediction time. MatchSense does not refit the baseline.

5.2 Query and fixture resolution

An LLM interpreter (by default `codex exec` with a strict JSON schema; then the OpenAI API; then a rule parser) returns intent, clubs, a *verbatim* date phrase and players in or out. It never returns dates or numbers. Clubs are validated against the canonical registry, and the date phrase is re-resolved deterministically in the user's IANA timezone. Because a 20:00 UK kickoff is 04:00

in Hong Kong, "tonight" is 17:00–06:00 local, and the resolved window is echoed back. Fixtures are never synthesised. If none falls in the window, the system says so and offers the next confirmed one. In-progress, finished and postponed fixtures, and competitions without a calibrated model (e.g. the UEFA Champions League), receive a status response rather than a probability.

5.3 Evidence extraction and corroboration

Items from the last 72 hours are grouped into syndication clusters when they share a canonical URL, have word-trigram Jaccard similarity ≥ 0.5 (Broder, 1997), or have title-token overlap ≥ 0.85 within 48 h. Each cluster counts once. An extractor maps each story to typed events with fields *entity*, *team*, *event type*, *certainty* $\in$ {confirmed, reported, speculation, AI inference}, *availability* $\in$ {available, doubtful, unavailable, unknown}, relevance and a supporting quote. The schema has no probability fields, and a test enforces this. Votes combine into a belief $\pi_i \in [0, 1]$:

$$\pi_i = \frac{\sum_k w_k v_k}{\sum_k w_k}, \qquad w_k = c_{\text{source}} \cdot 2^{-\text{age}_k/36\,\text{h}} \cdot c_{\text{certainty}}, \tag{3}$$

where v_k comes from a fixed table indexed by (availability, certainty), for example unavailable+confirmed $\to 0.02$ and unavailable+reported $\to 0.12$, or from FPL's published chance-of-playing percentage. Certainty weights are 1.0/0.7/0.35/0.25. Source credibilities range from 1.0 (official line-up) through 0.85 (BBC, Guardian) to 0.4 (tabloids). The most recent confirmed statement dominates. Disagreements above 0.4 are flagged as conflicts. *These constants are editorial assumptions, not estimates* (Section 10.2).

5.4 Start-probability (selection) model

For each squad player, $p_i = P(\text{start}_i \mid \text{available})$ is a pooled logistic regression on features computed from the team's earlier matches only: the exponentially weighted start rate τ_i (half-life four matches), started last match, starts in the last three, weighted minutes share, a short-rest indicator (≤ 4 days) and its interaction with "started last", log appearances, a goalkeeper flag and a never-started flag. C was chosen from $\{0.03, 0.1, 0.3, 1, 3\}$ on matchdays 16–22. The live start probability is $p_i \pi_i$. XIs are drawn by perturbing log-odds with Gumbel noise (Kool et al., 2019) and greedily choosing one goalkeeper and ten outfielders, subject to line caps of the team's usual shape +1. An official XI, once published, replaces the sample entirely.

5.5 Line-up impact model

Let $s_i \in \{0, 1\}$ indicate that player i starts. For team t facing o, define

$$D_t = 1 - \frac{\sum_i \tau_i s_i}{\sum_{\text{top-11}} \tau_i} \qquad \text{(discontinuity: usual starting mass absent)}, \tag{4}$$

$$G_t = \textstyle\sum_i g_i (1 - s_i) \qquad \text{(absent share of the team's recent goals)}, \tag{5}$$

$$D_t^{\text{def}} = 1 - \frac{\sum_{i \in \mathcal{D}} \tau_i s_i}{\sum_{\text{top-5},\, \mathcal{D}} \tau_i} \qquad \text{(goalkeeper + back four absent)}, \tag{6}$$

where g_i is player i's share of the team's decayed recent goals and $\mathcal{D}$ is the set of goalkeepers and defenders. The scoring rate of t is an offset Poisson GLM on the baseline:

$$\log \mathbb{E}[y_t] = \log \lambda_t^{\text{base}} + b_1 \tilde{D}_t + b_2 \tilde{D}_o + b_3 \tilde{G}_t + b_4 \tilde{G}_o + b_5 \tilde{D}_o^{\text{def}} \left[+ \textstyle\sum_i a_i (s_i - \tau_i) + \sum_j d_j (s_j - \tau_j) \right], \tag{7}$$

where tildes denote centring at training means, so a typical amount of rotation leaves the baseline unchanged. The bracketed player-level terms, a plus-minus analogue, appear only in some candidates. Coefficients are MAP estimates under Gaussian priors (ridge penalties). The Laplace covariance is $\Sigma = (X^\top \operatorname{diag}(\hat{\mu}) X + \Lambda)^{-1}$ (Tierney and Kadane, 1986).

Provisional prior. The deployed model uses informative priors $\boldsymbol{b} \sim \mathcal{N}(\boldsymbol{m}, \mathrm{diag}(\boldsymbol{s}^2))$ with $\boldsymbol{m} = (-0.15, +0.10, -0.25, 0, +0.20)$ and $\boldsymbol{s} = (0.15, 0.15, 0.15, 0.10, 0.15)$. These encode domain assumptions. Losing all of a side's recent scorers lowers its scoring rate by about 22%, and a fully changed XI lowers it by about 14%. A rotated opponent concedes slightly more, and an opponent missing its whole usual defensive unit concedes about 22% more. These are assumptions, updated by the 2025-26 likelihood.

5.6 Monte Carlo forecast and explanations

For $n = 1, \ldots, N$ ($N = 2000$): sample both XIs; sample $\boldsymbol{b}^{(n)} \sim \mathcal{N}(\hat{\boldsymbol{b}}, \Sigma)$; set $\lambda^{(n)} = \lambda^{\mathrm{base}} \exp(\Delta\eta^{(n)})$; and form the grid $P^{(n)}$. The forecast is $\bar{P} = N^{-1} \sum_n P^{(n)}$, so $P(H) + P(D) + P(A) = 1$ exactly. Uncertainty is reported as the 5–95% range of per-draw win probabilities and the Monte Carlo standard error. *Player impacts* are computed by setting one likely starter's availability to zero with common random numbers and recomputing the expected outcome. A *what-if* rebuilds both line-up models with the named players forced in or out, discards the official XI of an affected side, and reports the change in win probability and the likeliest replacements. All attributions are labelled as model associations, not causal effects (Hernán and Robins, 2020).

6 Experimental design

Information sets. (B) baseline only; (C) baseline + official XI, public at about $T - 60$ min; (P) baseline + an XI predicted from earlier selections only.

Baseline protocol. Expanding matchweek walk-forward on 2024-25 (validation) and 2025-26 (untouched test). Every fold is fitted only on matches dated before its earliest fixture. Temperatures and the Poisson–logistic blend weight $P = (1 - w)P_{\mathrm{Pois}} + wP_{\mathrm{logit}}$ are fitted on validation predictions only.

Line-up protocol. Matchday walk-forward on 2025-26. Fold k trains on every eligible observation that kicked off strictly before the earliest kickoff of matchday k, which also handles rearranged fixtures. Validation folds are matchdays 8–22 (150 matches) and test folds matchdays 23–38 (160 matches). Baseline rates come from the `ScoreModel` trained through 25 May 2025.

Metrics. Log loss (primary), multiclass Brier score, RPS, accuracy, exact-score NLL, reliability tables and paired bootstrap 95% intervals (4,000 resamples) for per-match log-loss differences.

Leakage controls (Kaufman et al., 2012). (i) Every fold asserts that its last training kickoff precedes the fold's first kickoff. (ii) Line-up state is snapshotted before a match is applied; a test rewrites all future line-ups, goals and results and asserts bit-identical earlier features. (iii) The baseline roll-forward ignores results finishing after the prediction time; a test mutates such a result and asserts identical probabilities. (iv) News published after the prediction time is excluded; a test verifies that a later confirmation changes the belief only once the clock passes its timestamp.

7 Results

7.1 Baseline: Poisson versus logistic

The goal-based model clearly beats the uninformed prior. The logistic alternative is slightly worse on both seasons, consistent with Hubáček et al. (2019). The 2025-26 season is harder than 2024-25, with a log-loss gap of 0.053.

	Validation 2024-25			Test 2025-26 (untouched)		
Model	Log loss	RPS	Brier	Log loss	RPS	Brier
Class-frequency prior	1.0852	0.2369	0.6586	1.0818	0.2278	0.6551
Poisson (`ScoreModel`)	0.9750	0.1989	0.5818	1.0282	0.2087	0.6179
Multinomial logistic	0.9846	0.2013	0.5886	1.0327	0.2096	0.6207
Poisson, temperature-scaled	**0.9748**	**0.1988**	**0.5814**	**1.0273**	**0.2086**	**0.6172**
Logistic, temperature-scaled	0.9824	0.2008	0.5868	1.0285	0.2088	0.6180

Table 2: Walk-forward baseline comparison (380 matches per season). The validation-selected blend weight on the logistic model is 0%. Temperatures: Poisson 1.051, logistic 1.149. Draw recall is 0 for every model: a draw is never the arg-max outcome. On the 95 test-season upsets (lower-Elo side wins), the Poisson model's log loss (1.29) is worse than the class prior's (1.06).

7.2 Start-probability model

Model (test: matchdays 23–38, 8,493 player-match rows)	Log loss	Top-11 precision
Typical start-rate heuristic	0.461	0.742
Pooled logistic, $C = 3$	**0.400**	**0.798**
Pooled logistic, normalised to exactly 11 starters	0.404	–

Table 3: Selection model. Validation log loss across C ranged from 0.442 ($C = 0.03$) to 0.432 ($C = 3$), against 0.514 for the heuristic. This is the one learned component with a clear, held-out gain.

7.3 Line-up impact candidates

Candidate (information set C)	Val. LL	Val. Brier	Test LL	Test Brier	Test RPS
Baseline (B)	1.0322	0.6220	1.0498	0.6338	0.2146
Continuity, $\alpha = 20$	1.0322	0.6220	1.0497	0.6335	0.2144
Continuity + goal share, $\alpha = 80$	1.0323	0.6221	1.0500	0.6338	0.2146
Continuity + defence, $\alpha = 5$ (one-SE choice)	1.0320	0.6217	1.0498	**0.6330**	**0.2140**
Continuity + defence, $\alpha = 20$	1.0319	0.6218	**1.0495**	0.6331	0.2142
Player effects, $\alpha = 20, \alpha_p = 100$ (best val.)	**1.0283**	**0.6189**	1.0560	0.6371	0.2160
Player effects, $\alpha = 5, \alpha_p = 30$	1.0296	0.6202	1.0765	0.6492	0.2218
Player effects, $\alpha = 20, \alpha_p = 10$	1.0451	0.6329	1.1369	0.6846	0.2386
Provisional prior (**deployed**)	1.0295	0.6205	1.0517	0.6349	0.2148

Table 4: Selected rows from the 22-candidate walk-forward (round 3). Full table in `artifacts/matchsense/REPORT.md`. Validation and test rankings barely agree. Player-level models win validation and lose test, and the loss grows as the player penalty α_p shrinks.

7.4 Posterior of the deployed layer

7.5 Does line-up information help at all?

We hid a random share of players' line-up indicators on the test folds, replacing them by typical start rates.

7.6 Calibration

Top-1 accuracy on the test folds is 0.469 (baseline) and 0.481 (provisional prior). This difference is not meaningful on 160 matches. The actual draw rate is 28.1%; neither model ever has draw as its arg-max.

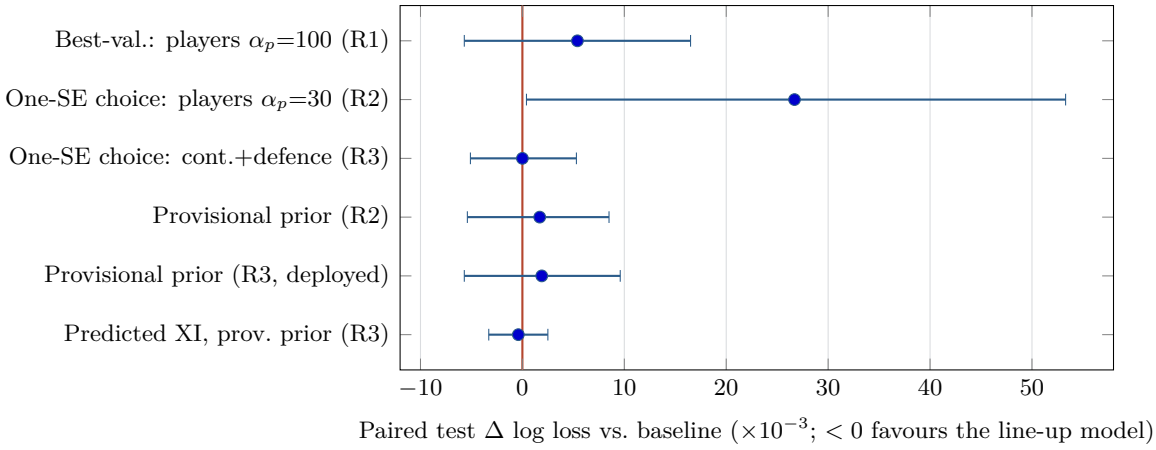


Figure 2: Held-out test effect of each round's chosen model, with 95% paired bootstrap intervals over 160 test matches. Only one interval excludes zero, and it lies on the *wrong* side (round 2, player effects). R1–R3: evaluation rounds (Section 7.7).

	Coefficient	Prior mean (sd)	Posterior mean ± sd
b_1	own discontinuity	−0.15 (0.15)	-0.150 ± 0.137
b_2	opponent discontinuity	+0.10 (0.15)	-0.009 ± 0.138
b_3	own absent goal share	−0.25 (0.15)	-0.187 ± 0.116
b_4	opponent absent goal share	0.00 (0.10)	-0.008 ± 0.087
b_5	opponent defensive discontinuity	+0.20 (0.15)	$+0.046 \pm 0.126$

Table 5: Deployed coefficients (log-rate scale). Posterior sds are barely smaller than prior sds, so the data are weakly informative. Where the data do move the posterior (b_2, b_5), they move it *towards zero*, against the domain prior.

7.7 Evaluation history and disclosure

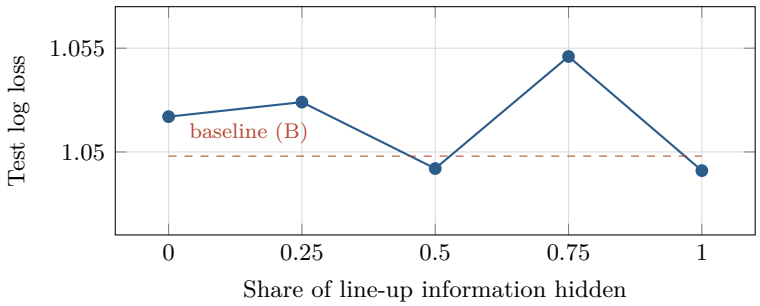


Figure 3: No monotone trend. Removing all line-up information is as good as keeping it, so in this sample line-up information has no measurable predictive value at the match-outcome level.

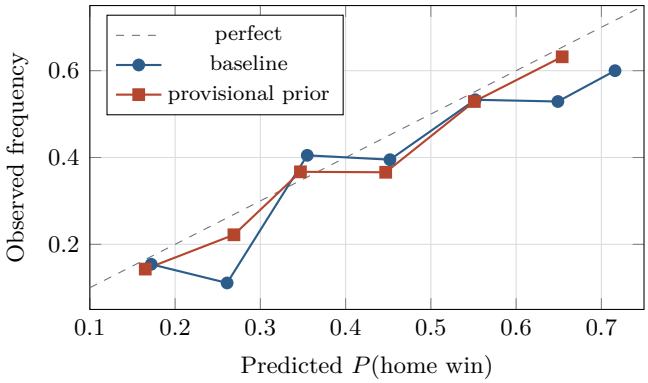


Figure 4: Reliability of home-win probabilities on the 160 test matches (bins with $n \geq 5$; bin sizes 5–41). Both are roughly calibrated, with a slight over-prediction of home wins above 0.6 for the baseline. Bins are too small for strong claims.

Round	Change	Outcome on test
R1	Select minimum-validation candidate	Player effects (α_p=100): +0.0054 [−0.0057, +0.0165]. Not an improvement.
R2	*After viewing R1 test*: one-SE rule, goal-share feature, provisional prior	One-SE choice (players, α_p=30) +0.0267 [+0.0004, +0.0533]: significantly worse. Prior deployed: +0.0017 [−0.0054, +0.0085].
R3	*After R2 and a user request about individual players*: defensive-discontinuity feature, 5-dim prior	One-SE choice (continuity+defence) +0.0000 [−0.0051, +0.0053]; deployed prior +0.0019 [−0.0057, +0.0096].

Table 6: Each round was designed after seeing the previous round's test results. The test folds are therefore **not an untouched holdout** for the deployed configuration (Cawley and Talbot, 2010). The neutral R3 result is a sanity check, not a confirmatory estimate.

8 Case study (demonstration data)

The demo replays a recorded FPL snapshot (4 Nov 2026) for Nottingham Forest v. Manchester City (kickoff Sun 8 Nov, 01:30 HKT), with **simulated** news from fictional publishers on `.invalid` domains. The figures below were regenerated with the round-3 artifacts for this report.

Demo clock (UTC)	System output
Wed 4 Nov 12:00	"No confirmed fixture tonight (17:00–06:00 HKT)"; offers Forest v City on 8 Nov. No probability is invented.
Sat 7 Nov 08:00	Haaland's corroborated availability $\pi = 0.31$ (doubtful; wire "doubt" + tabloid "out" and its syndicated copy merged). City: **52%** win / 24% draw / 24% loss; baseline alone 55%. Top contribution: Haaland availability, −0.10 City xG.
Same, what-if	"Without Haaland" vs. "with": City win −3.7 points. Likeliest replacements: Bouaddi (+0.21 start prob.), Foden (+0.11), Cherki (+0.10).
Sat 7 Nov 16:45	Simulated official XIs ($T - 60$ min) replace predictions; Haaland starts ($\pi = 0.98$). City **56%**. Haaland's modelled absence cost: −5.4 win points, −0.23 xG.

Table 7: Demo replay. All news is simulated; the numbers illustrate mechanics, not accuracy.

The replay also exposes model artefacts, discussed in Section 10.2. City's goalkeeper's absence lowers *City's own* xG (−0.10) far more than it raises Forest's (+0.01). A Forest defender carries a 29% share of Forest's recent goals and so ranks as Forest's most valuable starter.

9 Software validation

The repository's test suite (88 tests, all passing in about two minutes, offline) includes targeted MatchSense tests for alias resolution and timezone windows, "tonight" with no fixture, conflicting and syndicated reports, precedence of confirmed statements, the no-probability LLM contract, eleven-player XI validity, official-XI override, ambiguous player names, probabilities summing to one, missing credentials, cache retry and stale fallback, in-progress and unsupported competitions, and the four leakage tests of Section 6. The tests check engineering invariants. They cannot check forecasting skill.

10 Critical appraisal

10.1 Strengths

S1. No LLM-generated probabilities. The LLM's outputs are typed, validated and cached, and every LLM stage has a deterministic fallback. This removes the most common failure of "AI predictor" products.

S2. Honest reporting. The baseline is always displayed. The line-up layer is labelled provisional in every response. Negative results and post-hoc changes are documented in the repository.

S3. Leakage discipline. Strict time cut-offs, snapshotting and mutation tests at four points, plus fold audits in the artifacts.

S4. A validated baseline. The Poisson model beats a class-frequency prior by 0.054 log loss on an untouched season and is not improved by a logistic alternative.

S5. A useful selection model. Start probabilities improve log loss by 13% relative over a heuristic.

S6. Evidence provenance. Every belief links to sourced, timestamped, de-duplicated items. Conflicts are surfaced, not averaged away silently.

S7. Safe refusals. The system answers "no fixture", "match in progress" and "competition not modelled" rather than inventing numbers. It degrades gracefully without API keys.

S8. Coherent uncertainty. The Monte Carlo propagates both line-up and coefficient uncertainty, and probabilities sum to one by construction.

10.2 Flaws and weaknesses

Table 8 lists concrete flaws, each with a severity for the system's stated purpose of answering "will X win?" with news awareness.

10.3 Pros and cons versus simpler alternatives

10.4 Threats to validity

Internal. One season; post-hoc design changes (F2); availability constants are assumptions (F4); test-fold bins are small.

Construct. Discontinuity and goal share are crude proxies for player quality; defensive value is barely represented; positions are coarse.

External. One league with a stable top tier. The model refuses other competitions, but within-league regime changes (new managers, promoted sides) are only partly captured by the baseline's state transitions.

Conclusion. With 160 test matches, a paired log-loss interval has a half-width of about 0.005–0.01. Effects smaller than this, which is plausibly where real line-up effects lie, are undetectable.

10.5 Responsible use

Football forecasts are commonly used for betting. MatchSense's probabilities are roughly calibrated but carry no demonstrated edge, and the news layer is explicitly unvalidated. The system should not be used as a basis for wagering. The demo's news is simulated and must never be cited as fact. Only headline, summary, link and time are used from RSS feeds, and pages that forbid scraping are not fetched.

Flaw	Evidence and consequence	Severity
F1. News channel has no demonstrated value	Line-up effects are not identified (Figs. 2 and 3), and news enters *only* through line-ups. The headline feature, "news-aware", moves probabilities by priors, not by measured effects.	**High**
F2. Contaminated test folds	Three rounds were designed after viewing test results (Table 6). No clean estimate of the deployed model's skill exists.	**High**
F3. One season of line-up data	700 team-match observations against football's high outcome variance. Player effects are unidentifiable (Kharrat et al., 2020).	**High**
F4. Hand-set evidence constants	The availability table, certainty weights, credibilities and 36 h half-life are editorial. Some cells are counter-intuitive ("doubtful+speculation" $\rightarrow$ 0.6 is *more* available than "doubtful+confirmed" $\rightarrow$ 0.5). None is calibrated.	Medium
F5. Mis-attributed positional effects	The goalkeeper's absence mostly lowers his own team's attack (Section 8). Team-level features cannot distinguish roles, and b_5 was pulled towards zero.	Medium
F6. Noisy goal shares	Short decayed windows give defenders 29% goal shares; set pieces and own goals are mixed with open play. The "most valuable player" ranking is unstable.	Medium
F7. Draws never predicted	Draw recall is 0 in every model (28% of matches). Probabilities are fine, but the top-1 headline can never say "draw".	Medium
F8. Weak on upsets	Upset log loss 1.29 vs. 1.06 for the prior: the model is over-confident in favourites when they lose.	Medium
F9. Train/serve source shift	The selection model is trained on openfootball line-ups, but live features come from FPL histories (differing minute and position conventions). This is untested.	Medium
F10. Associational "what-if"	Counterfactuals use coefficients confounded by rotation against weak opponents and fixture congestion. They are not causal effects (Hernán and Robins, 2020).	Medium
F11. Coarse positions	FPL classes (wingers are "MID") make formations and the defensive unit approximate. The code enforces "back line $\geq$ usual size", whereas the model doc says "≥ 3".	Low
F12. Approximate XI sampler	Greedy Gumbel top-k under caps is not exact conditional-Bernoulli sampling (Chen et al., 1994), so marginals do not exactly equal $p_i\pi_i$.	Low
F13. Fragile data supply	FPL endpoints are undocumented, free NewsAPI is delayed, and cup fixtures need a key; otherwise answers must caveat that cup matches cannot be ruled out.	Low
F14. Query cost and latency	A new question through Codex costs about 20k tokens and 15 s (repeats are cached). This is heavy for parsing a short sentence.	Low
F15. Static baseline	The `ScoreModel` is not refitted in-season; only its state rolls forward. Drift within 2026-27 is uncorrected.	Low
F16. Scope	Premier League only, pre-match only, no player transfer of effects across clubs, tactical news not quantified.	Low

Table 8: Flaws identified from the evaluation artifacts, a code review and a replay of the demo.

Alternative	Pros relative to MatchSense	Cons relative to MatchSense
Baseline `ScoreModel` alone	Same measured accuracy; simpler; no news pipeline or keys; nothing provisional.	Cannot answer "what if X is out?"; ignores confirmed line-ups; no evidence trail.
Ask an LLM directly	Trivial to build; fluent; reads any source.	Uncalibrated, unverifiable numbers; hallucinated fixtures and injuries (Ji et al., 2023); no backtest possible.
Bookmaker / betting-market odds	Aggregate private information; typically sharper than public models (Dixon and Coles, 1997).	Include a margin; no explanation; no what-if; access and terms restrictions.
Event-data player models (VAEP, xG)	Can identify player value with far more data (Decroos et al., 2019).	Need licensed event data; heavy engineering; still confounded for selection.

Table 9: Positioning. For pure accuracy, MatchSense currently equals its own baseline. Its distinct value lies in transparency, refusal behaviour, evidence tracing and scenario analysis.

11 Future work

1. **More line-up seasons** with hierarchical shrinkage of player effects (Baio and Blangiardo, 2010; Kharrat et al., 2020), ideally with event-level xG.
2. **Prospective news archiving.** Store every retrieved item with its timestamp so that the news channel itself can be evaluated without leakage in a future season.
3. **Calibrate the evidence constants** (F4) against FPL's own chance-of-playing outcomes, which are observable after each match.
4. **Role-aware impact features** (keeper, centre-back, striker) to fix F5, and an explicit draw-decision rule for the top-1 headline (F7).
5. **A pre-registered, untouched evaluation** on 2026-27 with the configuration frozen now, using the prediction store's pre-kickoff scoring.
6. **Cheaper query parsing**: a small local model or the rule parser first, with the LLM only when the rules fail.

12 Conclusion

MatchSense shows how to connect unstructured football news to a validated forecaster without letting a language model invent probabilities. Evidence is typed, sourced, time-stamped and corroborated, and it enters the model only through player availability and line-up simulation. The evaluation separates what works from what does not. The goal-based baseline and the start-probability model are validated out of sample. Line-up effects on goal rates are not identified from one season, and the most expressive specification overfits. The deployed news adjustment is a labelled, prior-dominated model whose held-out effect is indistinguishable from zero. The system's present value is transparency, safe refusal and scenario analysis, not accuracy beyond its baseline. Whether news adds accuracy remains an open question. The project's own infrastructure (timestamped evidence and pre-kickoff prediction scoring) is what can answer it on a future season.

References

Gianluca Baio and Marta Blangiardo. Bayesian hierarchical model for the prediction of football results. *Journal of Applied Statistics*, 37(2):253–264, 2010.

Glenn W. Brier. Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1):1–3, 1950.

Andrei Z. Broder. On the resemblance and containment of documents. In *Proceedings of Compression and Complexity of Sequences*, pages 21–29. IEEE, 1997.

Rory P. Bunker and Fadi Thabtah. A machine learning framework for sport result prediction. *Applied Computing and Informatics*, 15(1):27–33, 2019.

Gavin C. Cawley and Nicola L. C. Talbot. On over-fitting in model selection and subsequent selection bias in performance evaluation. *Journal of Machine Learning Research*, 11:2079–2107, 2010.

Xiang-Hui Chen, Arthur P. Dempster, and Jun S. Liu. Weighted finite population sampling to maximize entropy. *Biometrika*, 81(3):457–469, 1994.

Anthony C. Constantinou and Norman E. Fenton. Solving the problem of inadequate scoring rules for assessing probabilistic football forecast models. *Journal of Quantitative Analysis in Sports*, 8(1), 2012.

Tom Decroos, Lotte Bransen, Jan Van Haaren, and Jesse Davis. Actions speak louder than goals: Valuing player actions in soccer. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, pages 1851–1861, 2019.

Mark J. Dixon and Stuart G. Coles. Modelling association football scores and inefficiencies in the football betting market. *Journal of the Royal Statistical Society: Series C (Applied Statistics)*, 46(2):265–280, 1997.

Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman & Hall, New York, 1993.

Arpad E. Elo. *The Rating of Chessplayers, Past and Present*. Arco, New York, 1978.

Edward S. Epstein. A scoring system for probability forecasts of ranked categories. *Journal of Applied Meteorology*, 8(6):985–987, 1969.

Tilmann Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007.

Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, volume 70 of *PMLR*, pages 1321–1330, 2017.

Trevor Hastie, Robert Tibshirani, and Jerome Friedman. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction*. Springer, New York, 2nd edition, 2009.

Miguel A. Hernán and James M. Robins. *Causal Inference: What If*. Chapman & Hall/CRC, Boca Raton, 2020.

Arthur E. Hoerl and Robert W. Kennard. Ridge regression: Biased estimation for nonorthogonal problems. *Technometrics*, 12(1):55–67, 1970.

Ondřej Hubáček, Gustav Šourek, and Filip Železný. Learning to predict soccer results from relational data with gradient boosted trees. *Machine Learning*, 108(1):29–47, 2019.

Lars Magnus Hvattum and Halvard Arntzen. Using ELO ratings for match result prediction in association football. *International Journal of Forecasting*, 26(3):460–470, 2010.

Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Ye Jin Bang, Andrea Madotto, and Pascale Fung. Survey of hallucination in natural language generation. *ACM Computing Surveys*, 55(12):248:1–248:38, 2023.

Dimitris Karlis and Ioannis Ntzoufras. Analysis of sports data by using bivariate Poisson models. *Journal of the Royal Statistical Society: Series D (The Statistician)*, 52(3):381–393, 2003.

Shachar Kaufman, Saharon Rosset, Claudia Perlich, and Ori Stitelman. Leakage in data mining: Formulation, detection, and avoidance. *ACM Transactions on Knowledge Discovery from Data*, 6(4):15:1–15:21, 2012.

Tarak Kharrat, Ian G. McHale, and Javier López Peña. Plus–minus player ratings for soccer. *European Journal of Operational Research*, 283(2):726–736, 2020.

Wouter Kool, Herke van Hoof, and Max Welling. Stochastic beams and where to find them: The Gumbel-top-k trick for sampling sequences without replacement. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *PMLR*, pages 3499–3508, 2019.

Siem Jan Koopman and Rutger Lit. A dynamic bivariate Poisson model for analysing and forecasting match results in the English Premier League. *Journal of the Royal Statistical Society: Series A (Statistics in Society)*, 178(1):167–186, 2015.

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474, 2020.

M. J. Maher. Modelling association football scores. *Statistica Neerlandica*, 36(3):109–118, 1982.

Peter McCullagh and John A. Nelder. *Generalized Linear Models*. Chapman & Hall, London, 2nd edition, 1989.

openfootball contributors. openfootball/england: Football.txt data for the english premier league and football league. `https://github.com/openfootball/england`, 2026. Public domain. Accessed November 2026.

Håvard Rue and Øyvind Salvesen. Prediction and retrospective analysis of soccer matches in a league. *Journal of the Royal Statistical Society: Series D (The Statistician)*, 49(3):399–418, 2000.

Joseph Sill. Improved NBA adjusted +/- using regularization and out-of-sample testing. In *MIT Sloan Sports Analytics Conference*, 2010.

Leonard J. Tashman. Out-of-sample tests of forecasting accuracy: An analysis and review. *International Journal of Forecasting*, 16(4):437–450, 2000.

The Premier League. Fantasy Premier League public endpoints (bootstrap-static, fixtures, element-summary). `https://fantasy.premierleague.com/api/`, 2026. Undocumented public JSON endpoints. Accessed November 2026.

Luke Tierney and Joseph B. Kadane. Accurate approximations for posterior moments and marginal densities. *Journal of the American Statistical Association*, 81(393):82–86, 1986.

A Reproducibility

```
python -m src.scripts.train_season_simulator --source local   # baseline
python -m src.scripts.compare_match_outcomes ./england        # Table 2
python -m src.scripts.train_lineup_impact                     # Tables 3-6, Figs 2-4
python -m src.scripts.serve --demo                            # Section 8
python -m pytest                                              # Section 9
```

All commands are run with the project virtual environment (`./.venv/bin/python`). Artifacts: `artifacts/walk_forward_comparison/REPORT.md`; `artifacts/matchsense/{metrics.json, REPORT.md, REPORT_round1.md, REPORT_round2.md}`. Software: Python 3.15.0, scikit-learn 1.9.1, SciPy 1.18.1, NumPy 2.5.4, FastAPI 0.162. Seeds: bootstrap 7; missing-data masks 11; Monte Carlo 20261104.

B Data availability

openfootball/england is public domain. The FPL endpoints are public but undocumented. BBC Sport RSS is used for headline, summary, link and time only. The demonstration news and official line-ups are simulated.